

CHAPTER - 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Big Basket E-Commerce Database Management System is designed to efficiently manage customers, categories, suppliers, products, carts, orders, payments, reviews, and deliveries. Functional dependencies help identify the relationship between attributes in each table, while redundancy analysis and data anomaly detection ensure data consistency and reduce duplication. A well-designed database improves performance, maintains data integrity, and supports efficient order management.

1. CUSTOMER TABLE

Attribute	Key Type
Customer_ID	Primary Key
Customer Name	Not Null
Email	Unique Key
Phone No	Unique Key
Date of Birth	-
Address	-

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_No, DOB, Address

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders, Carts, and Reviews store only the Customer_ID, preventing duplicate customer records.

Data Anomalies

- **Insert Anomaly:** A new customer can be added without placing an order.
- **Update Anomaly:** Updating a customer's phone number or address requires changes only in the Customer table.
- **Delete Anomaly:** Deleting an order does not remove customer information.

2. CATEGORY TABLE

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

Functional Dependency

Category_ID $\rightarrow$ Category_Name, Description

Redundancy Analysis

Each category is stored only once. Products reference the category using Category_ID instead of repeating the category name and description.

Data Anomalies

- **Insert Anomaly:** New categories can be added without any product being linked to them.
- **Update Anomaly:** Renaming a category or changing its description is done only once, in the Category table.
- **Delete Anomaly:** Deleting a product does not delete the category.

3. SUPPLIER TABLE

Attribute	Key Type
-----------	----------

Supplier_ID	Primary Key
Supplier Name	Not Null
Contact_No	Unique Key
Address	-

Functional Dependency

Supplier_ID → Supplier_Name, Contact_No, Address

Redundancy Analysis

Supplier details are stored only once in the Supplier table. Products reference the supplier using Supplier_ID instead of repeating supplier information.

Data Anomalies

- **Insert Anomaly:** A new supplier can be added before supplying any product.
- **Update Anomaly:** Supplier contact details are updated only once, in the Supplier table.
- **Delete Anomaly:** Deleting a product does not remove supplier information.

4. PRODUCT TABLE

Attribute	Key Type
Product_ID	Primary Key
Product Name	-
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Brand	-
Price	Check
Stock	Check
Expiry_date	-

Functional Dependency

Product_ID $\rightarrow$ Product_Name, Category_ID, Supplier_ID, Brand, Price, Stock, Expiry_Date

Redundancy Analysis

Product details are stored only in the Product table. Cart_Item, Order_Item, and Review use Product_ID instead of repeating product information.

Data Anomalies

- **Insert Anomaly:** New products can be added before receiving any orders.
- **Update Anomaly:** Product price or stock is updated only once, in the Product table.
- **Delete Anomaly:** Deleting an order does not remove product information.

5. CART TABLE

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_items	-
Total_amount	-

Functional Dependency

Cart_ID $\rightarrow$ Customer_ID, Total_Items, Total_Amount

Redundancy Analysis

The Cart table stores only cart-related information for a customer. Customer details are referenced using Customer_ID instead of being repeated.

Data Anomalies

- **Insert Anomaly:** A cart can be created for a customer independently of any order.

- **Update Anomaly:** Cart totals are updated only in the Cart table without affecting customer data.
- **Delete Anomaly:** Deleting a cart does not delete the customer's information.

6. CART_ITEM TABLE

Attribute	Key Type
Cart_Item_ID	Primary Key
Cart_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Check

Functional Dependency

$\text{Cart_Item_ID} \rightarrow \text{Cart_ID}, \text{Product_ID}, \text{Quantity}$

Redundancy Analysis

This table connects carts and products, avoiding repeated product information inside the Cart table.

Data Anomalies

- **Insert Anomaly:** Products can be added to a cart independently, referencing an existing Cart_ID and Product_ID.
- **Update Anomaly:** Quantity changes affect only the selected cart item.
- **Delete Anomaly:** Removing a cart item does not delete the product.

7. ORDERS TABLE

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-

Order_status	Check
Total_amount	-

Functional Dependency

Order_ID $\rightarrow$ Customer_ID, Order_Date, Order_Status, Total_Amount

Redundancy Analysis

The Order table stores only order-related information. Customer and product details are referenced using foreign keys.

Data Anomalies

- **Insert Anomaly:** Orders can be created independently using an existing Customer_ID.
- **Update Anomaly:** Order status can be updated without affecting customer or product tables.
- **Delete Anomaly:** Deleting an order does not remove customer or product data.

8. ORDER_ITEM TABLE

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Price	-

Functional Dependency

Order_Item_ID $\rightarrow$ Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

This table connects orders and products, avoiding repeated product information in the Order table.

Data Anomalies

- **Insert Anomaly:** Products can be added to an order independently, referencing an existing Order_ID and Product_ID.
- **Update Anomaly:** Quantity or price changes affect only that order item.
- **Delete Anomaly:** Deleting an order item does not remove the product.

9. PAYMENT TABLE

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_method	-
Payment_status	Check
Payment_date	-
Amount	-

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment information is stored separately from the Order table to avoid duplicate payment records.

Data Anomalies

- **Insert Anomaly:** Payment details can be recorded after an order is created.
- **Update Anomaly:** Payment status is updated only once, in the Payment table.
- **Delete Anomaly:** Deleting a payment record does not remove the order.

10. REVIEW TABLE

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	
Review_text	-
Review_date	-

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Reviews store only references to Customer_ID and Product_ID rather than duplicating customer or product information.

Data Anomalies

- **Insert Anomaly:** A customer can submit a review after purchasing a product.
- **Update Anomaly:** Rating or review text is updated only in the Review table.
- **Delete Anomaly:** Deleting a review does not delete the product or customer.

11. DELIVERY TABLE

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_address	-
Delivery_date	-
Delivery_status	Check

Functional Dependency

$\text{Delivery_ID} \rightarrow \text{Order_ID}, \text{Delivery_Address}, \text{Delivery_Date}, \text{Delivery_Status}$

Redundancy Analysis

Delivery details are maintained in a separate table, preventing repeated order information.

Data Anomalies

- **Insert Anomaly:** Delivery records can be added after an order is confirmed.
- **Update Anomaly:** Delivery status can be updated independently, in the Delivery table.
- **Delete Anomaly:** Deleting delivery information does not remove the order.

CONCLUSION

The Big Basket E-Commerce Database System is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains integrity, and ensures efficient database management for the e-commerce system.